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ggret: An R package for visualising and manipulating tree-based phylogenetic networks

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Summary

Evolutionary relationships of biological entities are most often modeled with phylogenetic trees. Phylogenetic trees consist of branches (or edges) representing direct lines of descent or genetic flow from ancestor to offspring (*i.e.* lineages), and nodes representing evolutionary "splits" through which a parental lineage gives rise to multiple child lineages. This vertical model of evolution has provided immense insights into the evolutionary history and processes underlying observed biological diversity. However, it fails to account for "horizontal" modes of evolution, whereby genetic material can be exchanged between contemporaneous organisms through a variety of mechanisms across the tree of life (Arnold et al., 2022; Keeling, 2024; Pérez-Losada et al., 2015). In recent years, advances in sequencing technologies and computational methods have made it increasingly possible to integrate horizontal evolutionary events into reticulated phylogenetic trees (or phylogenetic networks; (Chen & Wang, 2010; Huson & Bryant, 2006; Müller et al., 2020, 2022; Vaughan et al., 2017)). While phylogenetic networks have the potential to provide more comprehensive and accurate evolutionary pictures for many biological groups, the development of specific tools is required for their manipulation and visualisation. Here, we present `ggret`, an R package building upon the popular `ggtree` package (Yu et al., 2017) for the manipulation and visualisation of phylogenetic networks. `ggret` provides novel functions for parsing extended Newick and NEXUS files and introduces the `geom_ret` object for visualisation of tree-based phylogenetic networks.

Statement of need

Command-line tools are needed for the efficient, reproducible manipulation and plotting of phylogenetic networks. The R language is commonly used for phylogenetic analysis and visualisation with packages such as ape and `ggtree` providing important functionalities for handling and visualizing classic phylogenetic trees (Paradis & Schliep, 2019; Yu et al., 2017). `ggret` extends `ggtree` with methods for manipulating and plotting tree-based phylogenetic networks, where horizontal transfers of genetic material are represented by reticulation edges connecting the branches of a backbone phylogenetic tree. `ggret` enables the parsing of phylogenetic networks in extended Newick and NEXUS formats such as those produced by the popular *BEAST2* software (Bouckaert, 2019). Importantly, it can handle associated metadata via `treedata` objects and maintains most `ggtree` annotation and visualisation methods.

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Usage

ggret is available on GitHub and can be installed by using `remotes' install_github` function.

```
# Install from GitHub
if (!requireNamespace("remotes", quietly=TRUE))
  install.packages("remotes")
remotes::install_github("grdspcht/ggret", dependencies = TRUE,
                        build_vignettes = TRUE)
```

ggret provides functions for reading extended Newick and (*BEAST2*) NEXUS format (Cardona et al., 2008). This allows the user to visualise phylogenetic networks inferred from various programs. Here we parsed a (summary) phylogenetic network simulated with the *BEAST2* package *Bacter* (Vaughan et al., 2017) containing various node metadata, using the `read_beast_retnet` function. Note that the resulting retnet `treedata` object has been included in the package for the sake of reproducibility.

```
retfile <- system.file("extdata", "retnet.nexus", package = "ggret")
retnet <- ggret::read_beast_retnet(retfile)
```

ggret is the central function of this package. In a simple call without additional arguments it plots a rudimentary tree-based network without labeling. Reticulated edges are drawn as black dashed lines by default, but ggret contains various arguments to change their aspects (e.g. colour, linetype, etc.) (Figure 1).

```
library(ggret)

# Simple network
p1 <- ggret::ggret(retnet_treedata)
# Reticulation edges displayed as red dotted lines, in a "snake" shape
p2 <- ggret::ggret(retnet_treedata, retcol = "red", retlinetype = 3)
# Reticulation edges displayed as blue solid lines, in a straight shape and
# with arrow heads
p3 <- ggret::ggret(retnet_treedata, retcol = "blue", retlinetype = 1, arrows = T,
                   rettype = "straight")
```

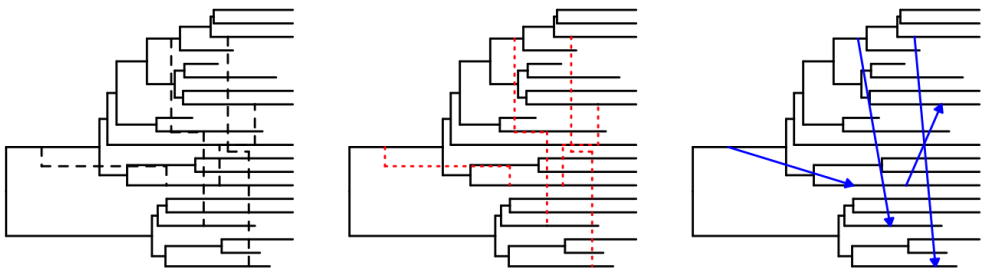

Figure 1: A simple tree-based networks plotted with ggret using different formats for reticulation edges.

We can rotate some of the nodes to avoid crossing of reticulation edges and improve visualisation using the `ggtree rotate` function. Note that one can initially visualise node indices using `ggret(retnet) + geom_nodelab(aes(label=node))` to make this easier (Figure 2).

```
# Rotate nodes to improve vizualisation
ggtree::rotate(p1, node = 31) %>%
  ggtree::rotate(node = 30) %>%
  ggtree::rotate(node = 26) %>%
```

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```
ggtree::rotate(node = 37) ->
p1
```

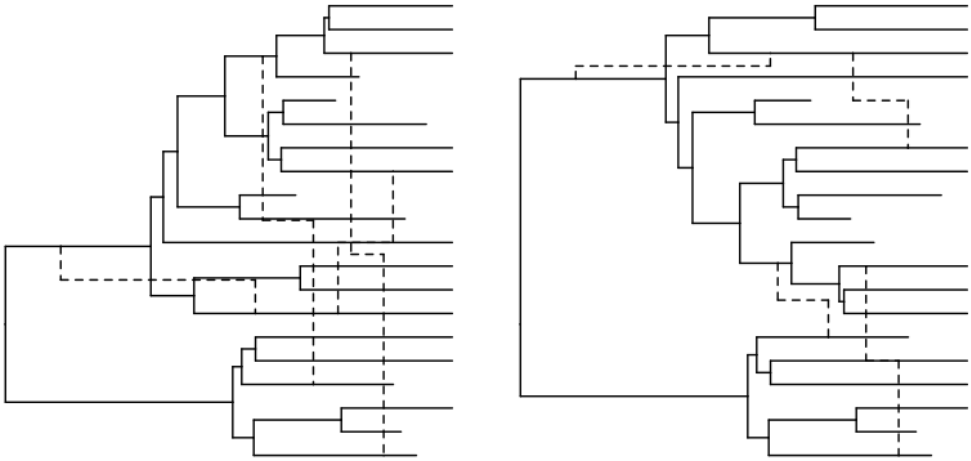

Figure 2: An example network before rotation (left) and the same network plotted after node rotation (right) to limit the number of edge crossings.

Annotations can be added using `ggtree` functions, such as `geom_tiplab`, `geom_nodelab` and `geom_range`. In addition, a time axis may be added by using the `theme_tree2` theme (Figure 3).

```
# Get the tMRCA of the tree and define time points to display
# the time axis in years BP
tmrca <- phytools::nodeHeights(retnet_treedata@phylo) %>% max
xticks_BP <- c(20000, 15000, 10000, 5000, 0)

# Add tip and node labels (we expand the x axis limits
# so that tip labels still fit).
p1 <- p1 +
  ggtree::geom_tiplab() +
  ggtree::geom_nodelab(aes(label = round(posterior, 2)),
          vjust = -0.25, hjust = 1.3, size = 3) +
  ggplot2::expand_limits(x = 22000)

# Adding a time axis and node bars indicating node heights' 95%
# highest posterior density interval.
p1 <- p1 +
  ggtree::theme_tree2() +
  ggtree::geom_range(aes(range = "height_95_HPD"), color = "grey50",
          alpha = .6, size = 1.5) +
  ggtree::scale_x_continuous(breaks = tmrca - xticks_BP, labels = xticks_BP) +
  ggplot2::xlab("Years before present")
```

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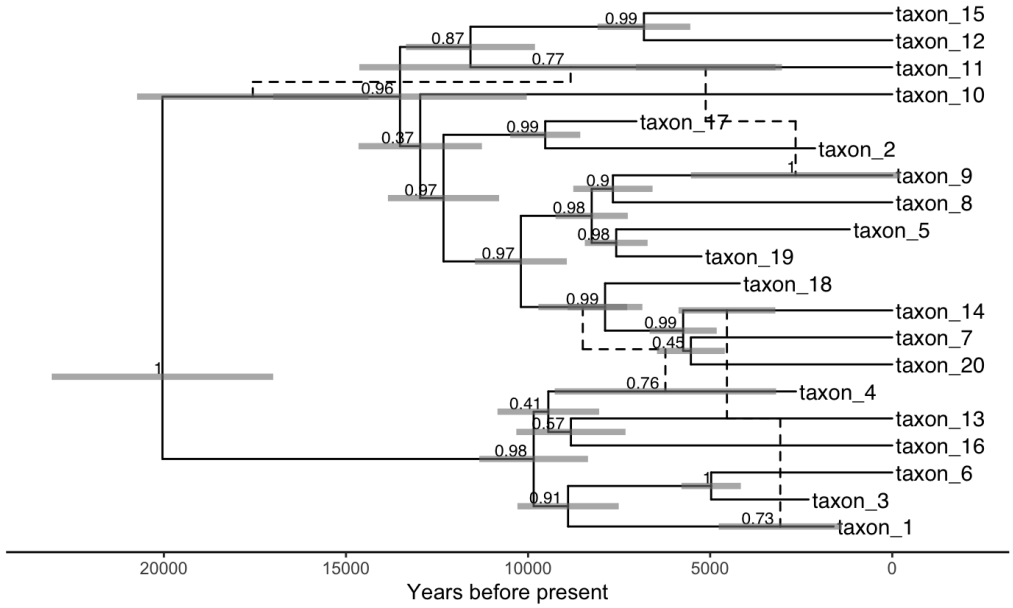

Figure 3: Annotated phylogenetic network. Tip labels are plotted and internal nodes are labelled with their posterior probability. The x-axis represents a timeline in years before present. The 95% highest probability density intervals of nodes' ages are indicated by grey bars.

The `group_clade` can be used to define clades within a network and color them accordingly. `group_clade` assigns clade information to all nodes descending from the MRCA of tips specified in the `nodes` argument (Figure 4).

```
# Define clades using the groupClade function
retnet_treedata %>%
  ggret::group_clade(nodes = c("taxon_10", "taxon_20"), cladename = "A",
         addtotreedata = T) %>%
  ggret::group_clade(nodes = c("taxon_11", "taxon_15"), cladename = "B",
         addtotreedata = T) %>%
  ggret::group_clade(nodes = c("taxon_1", "taxon_16"), cladename = "C",
         addtotreedata = T) ->
  retnet_clade

# Plot network with colored clades
ggret::ggret(retnet_clade, aes(color = clade))
```

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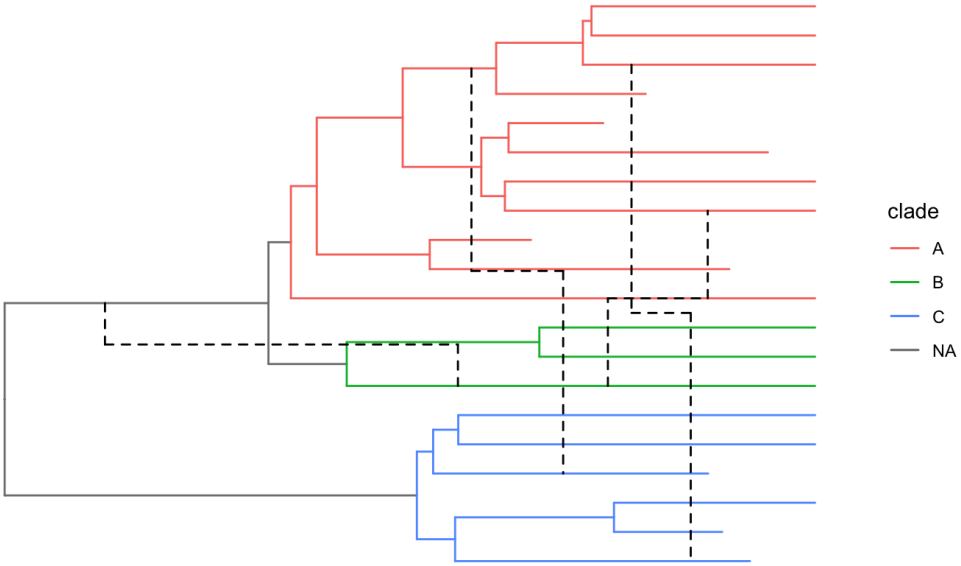

Figure 4: Phylogenetic network with colored based on clade information. Deep branches not belonging to any defined clade are labelled as NA.

For additional information, refer to `ggret`'s internal documentation or to https://github.com/grdspcht/ggret which also gives users the opportunity to open issues, pull requests or report bugs.

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References

Arnold, B. J., Huang, I.-T., & Hanage, W. P. (2022). Horizontal gene transfer and adaptive evolution in bacteria. *Nature Reviews Microbiology*, *20*(4), 206–218. https://doi.org/10.1038/s41579-021-00650-4

Bouckaert, T. G. A. B.-S., Remco AND Vaughan. (2019). BEAST 2.5: An advanced software platform for bayesian evolutionary analysis. *PLOS Computational Biology*, *15*(4), 1–28. https://doi.org/10.1371/journal.pcbi.1006650

Cardona, G., Rosselló, F., & Valiente, G. (2008). Extended Newick: It is time for a standard representation of phylogenetic networks. *BMC Bioinformatics*, *9*(1), 532. https://doi.org/10.1186/1471-2105-9-532

Chen, Z.-Z., & Wang, L. (2010). HybridNET: A tool for constructing hybridization networks. *Bioinformatics (Oxford, England)*, *26*(22), 2912–2913. https://doi.org/10.1093/bioinformatics/btq548

Huson, D. H., & Bryant, D. (2006). Application of phylogenetic networks in evolutionary studies. *Molecular Biology and Evolution*, *23*(2), 254–267. https://doi.org/10.1093/molbev/msj030

Keeling, P. J. (2024). Horizontal gene transfer in eukaryotes: Aligning theory with data. *Nature Reviews Genetics*, 1–15. https://doi.org/10.1038/s41576-023-00688-5

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Müller, N. F., Kistler, K. E., & Bedford, T. (2022). A bayesian approach to infer recombination patterns in coronaviruses. *Nature Communications*, *13*(1), 4186. https://doi.org/10.1038/s41467-022-31749-8

Müller, N. F., Stolz, U., Dudas, G., Stadler, T., & Vaughan, T. G. (2020). Bayesian inference of reassortment networks reveals fitness benefits of reassortment in human influenza viruses. *Proceedings of the National Academy of Sciences*, *117*(29), 17104–17111. https://doi.org/10.1073/pnas.1918304117

Paradis, E., & Schliep, K. (2019). Ape 5.0: An environment for modern phylogenetics and evolutionary analyses in R. *Bioinformatics*, *35*(3), 526–528. https://doi.org/10.1093/bioinformatics/bty633

Pérez-Losada, M., Arenas, M., Galán, J. C., Palero, F., & González-Candelas, F. (2015). Recombination in viruses: Mechanisms, methods of study, and evolutionary consequences. *Infection, Genetics and Evolution*, *30*, 296–307. https://doi.org/10.1016/j.meegid.2014.12.022

Vaughan, T. G., Welch, D., Drummond, A. J., Biggs, P. J., George, T., & French, N. P. (2017). Inferring ancestral recombination graphs from bacterial genomic data. *Genetics*, *205*(2), 857–870. https://doi.org/10.1534/genetics.116.193425

Yu, G., Smith, D. K., Zhu, H., Guan, Y., & Lam, T. T.-Y. (2017). Ggtree: An R package for visualization and annotation of phylogenetic trees with their covariates and other associated data. *Methods in Ecology and Evolution*, *8*(1), 28–36. https://doi.org/10.1111/2041-210X.12628

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